

FINAL EXAMINATION

MA 109C: INTRODUCTION TO GEOMETRY AND TOPOLOGY

- (1) (a) State the definition of a smooth vector field v on a manifold X , and the definition of the index of v at an isolated zero $x \in X$ of v .
(b) Let X be a compact oriented manifold without boundary. State the definition of the Euler characteristic $\chi(X)$ of X . State the Poincaré–Hopf theorem for X .
(c) Prove that for positive integers $k \geq 1$,

$$\chi(\mathbb{S}^k) = \begin{cases} 2, & k \text{ even} \\ 0, & k \text{ odd.} \end{cases}$$

- (d) What is $\chi(SO(8))$, where $SO(8)$ is the special orthogonal group consisting of the 8×8 real orthogonal matrices of determinant 1?

- (2) In this problem, we will prove that there exist real numbers $x, y, z, w \in \mathbb{R}$ such that

$$\begin{cases} x^3 - 3xy^2 - 3xz^2 - 3xw^2 - 3625x^2 - 121 = 0, \\ 3x^2y - y^3 - yz^2 - yw^2 + 6495x - 74625w = 0, \\ 3x^2z - y^2z - z^3 - zw^2 + 143xy = 0, \\ 3x^2w - y^2w - z^2w - w^3 - 10 = 0. \end{cases} \quad (1)$$

The strategy of the proof will be to imitate the proof of the Fundamental Theorem of Algebra.

- (a) State the Fundamental Theorem of Algebra.

Following the proof of the Fundamental Theorem of Algebra, write the above system (1) of equations as

$$F(x, y, z, w) + G(x, y, z, w) = 0,$$

where $F, G : \mathbb{R}^4 \rightarrow \mathbb{R}^4$ are the polynomials such that each component of F is homogeneous of degree 3 and each component of G has degree less than 3. It follows from a computation, which you are not required to show, that

$$\begin{aligned} & (x^3 - 3xy^2 - 3xz^2 - 3xw^2)^2 + (3x^2y - y^3 - yz^2 - yw^2)^2 \\ & + (3x^2z - y^2z - z^3 - zw^2)^2 + (3x^2w - y^2w - z^2w - w^3)^2 \\ & = (x^2 + y^2 + z^2 + w^2)^3, \quad (x, y, z, w) \in \mathbb{R}^4, \end{aligned}$$

or equivalently,

$$|F(x, y, z, w)| = |(x, y, z, w)|^3, \quad (x, y, z, w) \in \mathbb{R}^4,$$

where $|\cdot|$ denotes the Euclidean norm in $\mathbb{R}^4$.

- (b) Show that there exists some $R > 0$ such that

$$|F(x, y, z, w)| > |G(x, y, z, w)|$$

whenever $(x, y, z, w) \in \mathbb{R}^4$ is such that $|(x, y, z, w)| = R$.

Now, assume for contradiction that there are no solutions to (1) in $\mathbb{R}^4$.

- (c) For $R > 0$ given as in (b), show that the map $F/|F| : \partial B_R(0) \rightarrow \mathbb{S}^3$ is well-defined and has degree zero.
 (d) Use the Borsuk–Ulam theorem on $F/|F| : \partial B_R(0) \rightarrow \mathbb{S}^3$ to deduce that the mod 2 degree of $F/|F|$ is 1. Show that its degree must be odd.

The contradiction between (c) and (d) tells us that (1) must have a solution, as we set out to prove.

- (3) (a) Let $M, N \subseteq \mathbb{R}^q$ be compact oriented submanifolds without boundaries, of dimensions m, n respectively. Assume that M and N are disjoint and $m + n = q - 1$. The *linking number* $\ell(M, N)$ is defined to be the degree of the map

$$M \times N \rightarrow \mathbb{S}^{q-1}, \quad (x, y) \mapsto (x - y)/|x - y|.$$

Prove that $\ell(N, M) = (-1)^{(m-1)(n-1)}\ell(M, N)$.

- (b) Show that if M is the boundary of an oriented submanifold $X \subseteq \mathbb{R}^q$, then $\ell(M, N) = 0$.
- (c) If $M, N \subseteq \mathbb{S}^q$ are compact oriented submanifolds without boundaries of dimensions m, n respectively such that M, N are disjoint and $m + n = q - 1$, one may take a point $c \in \mathbb{S}^q \setminus (M \cup N)$ and compute the linking number in $\mathbb{S}^q \setminus \{c\} = \mathbb{R}^q$. Argue why the choice of c is irrelevant for the linking number.
- (d) If $y \neq z$ are regular values for a map $f : \mathbb{S}^{2q-1} \rightarrow \mathbb{S}^q$, then the manifolds $f^{-1}(y)$ and $f^{-1}(z)$ can be oriented with the preimage orientation, hence the linking number $\ell(f^{-1}(y), f^{-1}(z))$ is defined. Show that this is locally constant as functions of y and z .
- (e) If $g : \mathbb{S}^{2q-1} \rightarrow \mathbb{S}^q$ is another function, if y, z are common regular values of f and g , and if

$$|f(x) - g(x)| < |y - z|, \quad \forall x \in \mathbb{S}^{2q-1},$$

prove that

$$\ell(f^{-1}(y), f^{-1}(z)) = \ell(g^{-1}(y), g^{-1}(z)) = \ell(g^{-1}(y), g^{-1}(z)).$$

- (f) Prove that $\ell(f^{-1}(y), f^{-1}(z))$ depends only on the homotopy class of f and does not depend on the choice of y and z . The well-defined integer $H(f) = \ell(f^{-1}(y), f^{-1}(z))$ is called the *Hopf invariant* of f .
- (g) If q is odd, show that for any $f : \mathbb{S}^{2q-1} \rightarrow \mathbb{S}^q$ we have $H(f) = 0$.
- (h) What is the Hopf invariant of a constant map $\mathbb{S}^{2p-1} \rightarrow \mathbb{S}^p$?
- (i) The *Hopf fibration* $\pi : \mathbb{S}^3 \rightarrow \mathbb{S}^2$ is defined by

$$\pi(x, y, z, w) = h^{-1} \left(\frac{x + iy}{z + iw} \right)$$

where h denotes stereographic projection onto the complex plane. Compute that $H(\pi) = 1$. Show that π is not homotopic to a constant map.